

Theorem: Any planar graph with n vertices can be 6-colored.

Proof by Induction (on # vertices, n)

- Base Case ($n \leq 6$): we color each vertex with a different color ✓

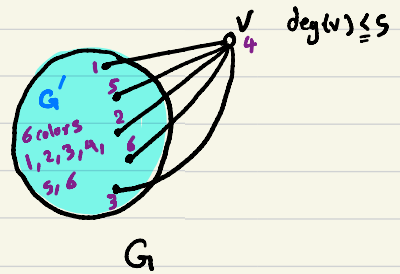
- I.H.: Any planar graph with $< n$ vertices can be 6-colored.

- I.S.: Any planar graph G with n vertices can be 6-colored.

By previous Thm. there is a vertex v of $\deg \leq 5$.

$$G' = G - v$$

G' has $n-1$ vertices.



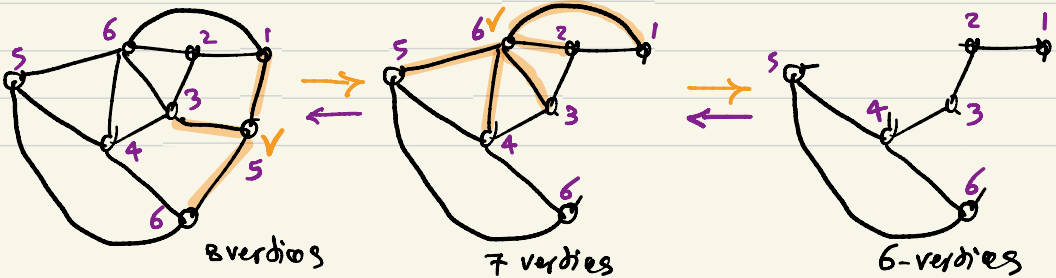
By I.H. G' can be 6-colored.

v has at most 5 neighbors in G'

These neighbors can use at most five colors

We can color v by the 6th color that is not used by its neighbors.

This gives a proper 6-coloring of G . □



Six-Coloring (G)

- if $|G| \leq 6$ then color the vertices 1, 2, 3, 4, 5, 6

- else

v : a vertex of $\deg. \leq 5$ in G

Six-Coloring($G-v$)

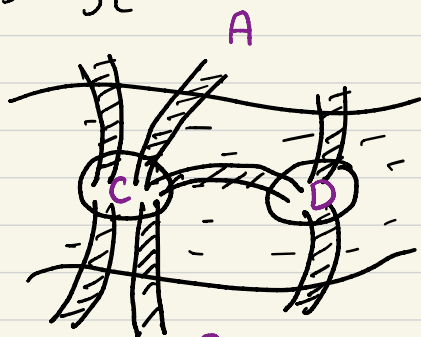
color v with one of 1, 2, 3, 4, 5, 6 that is not used by v 's neighbors.

end

end

Eulerian Graphs

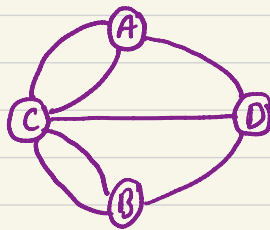
7-bridge



B

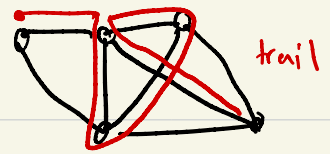
L. Euler (1736): NO

Is it possible to start somewhere and walk over every bridge exactly once?

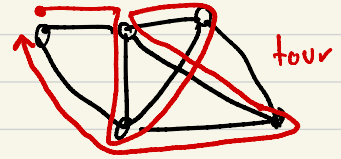


Is it possible to start from a vertex and walk on each edge exactly once?

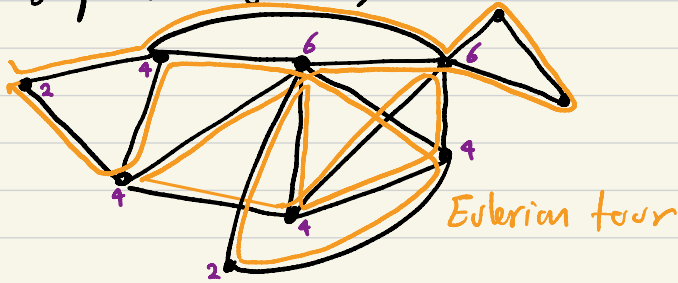
trail: a walk without repeating edges.



tour/circuit: a closed trail.



Euler tour/circuit: a tour that uses all edges of the graph (exactly once)



a graph is Eulerian if it has an Euler tour.

Theorem: a ^{connected} graph has an Euler tour if and only if all vertices $\xrightarrow{\hspace{1cm}}$ have even degrees.

Proof:

→ assume G has Euler tour

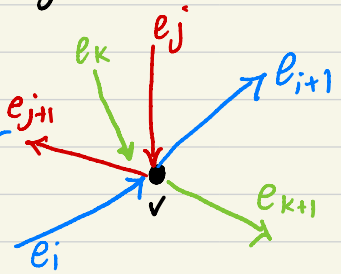
we show each vertex has even degree.

take an arbitrary vertex v

each time we visit v along the tour

we enter from an edge and exit

from a different edge.



thus $\deg(v)$ is twice the # times v is visited.

thus $\deg(v)$ is even.

← assume all vertices have even degrees

show G has an Euler tour

Consider longest (max # of edges) trail T in G .

$$T = v_0, v_1, v_2, \dots, v_{k-1}, v_k$$



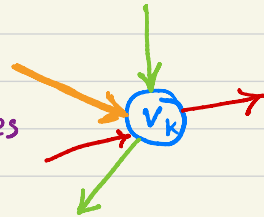
Since T is longest all edges incident to v_0 and v_k are used by T , because otherwise we could make T longer.

Claim: $v_0 = v_k$ (Claim that T is closed)

prove this claim by contradiction

assume $v_0 \neq v_k$

each time we visit v_k we use two edges except for the last time

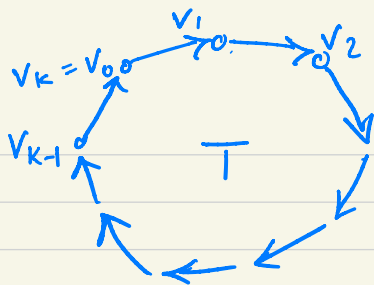


if we visit v_k say j times then the degree of v_k is $2j-1$

$2j-1$ is odd and this contradicts our assumption

that all vertices have even degrees $\square$

$\Rightarrow$ Therefore T is closed (is a tour)



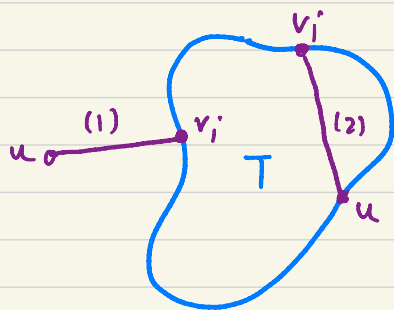
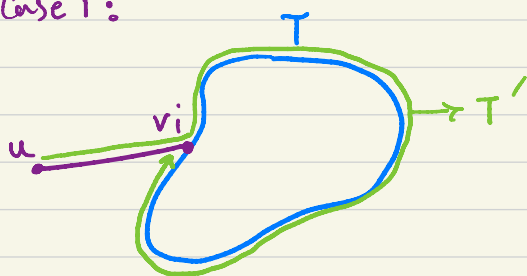
Claim: T is an Euler tour (uses all the edges)

we prove by contradiction

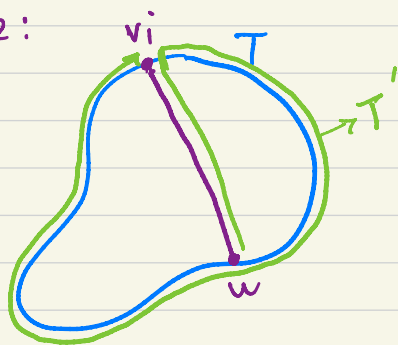
assume T does not use all edges

Since G is connected there must be an edge (u, v_i) that is not part of T .
(such an edge is connected to T)

Case 1:

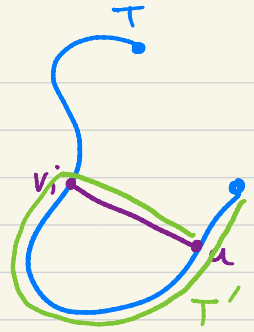
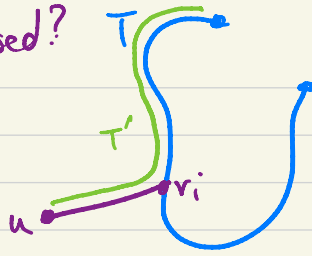


Case 2:



In both cases we get
a trail T' longer than T ,
this contradicts our choice of T
as the longest trail $\square$

What if T was not closed?



Eulerian trail: a trail that uses all edges of graph.

Thm: a connected graph has an Eulerian trail
if it has at most 2 vertices of odd degree

